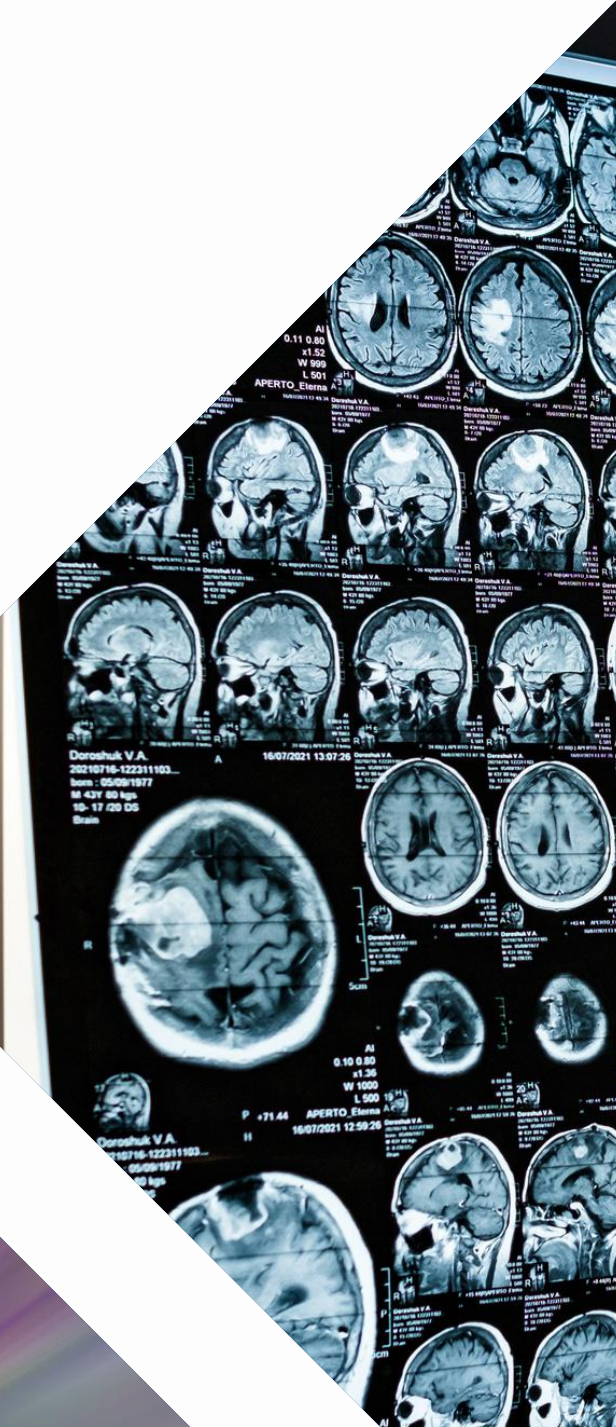


EPIXa – Epilepsy eXpected Alert

Smart System for Early Prediction of
Epileptic Seizures

SUPERVISION

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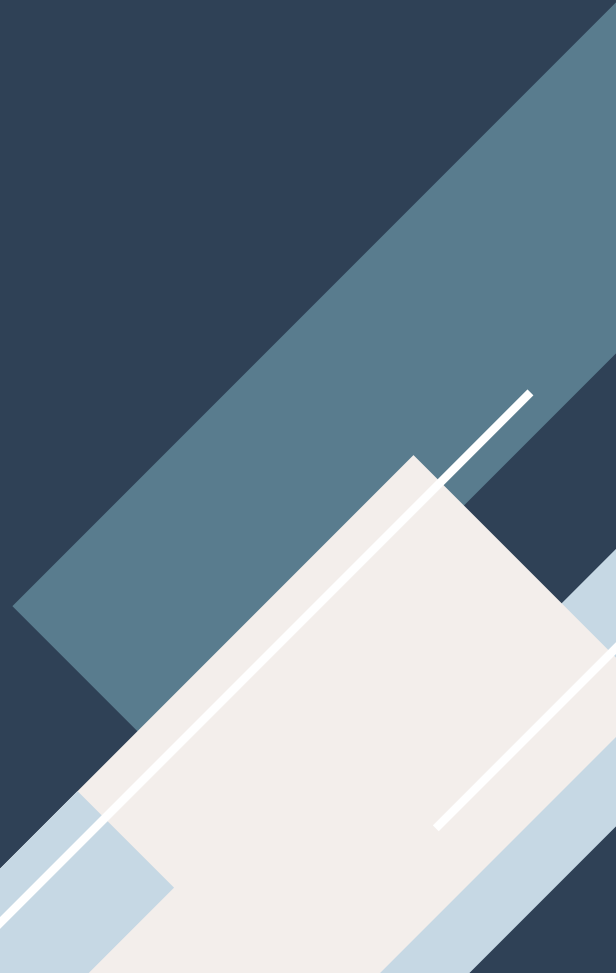
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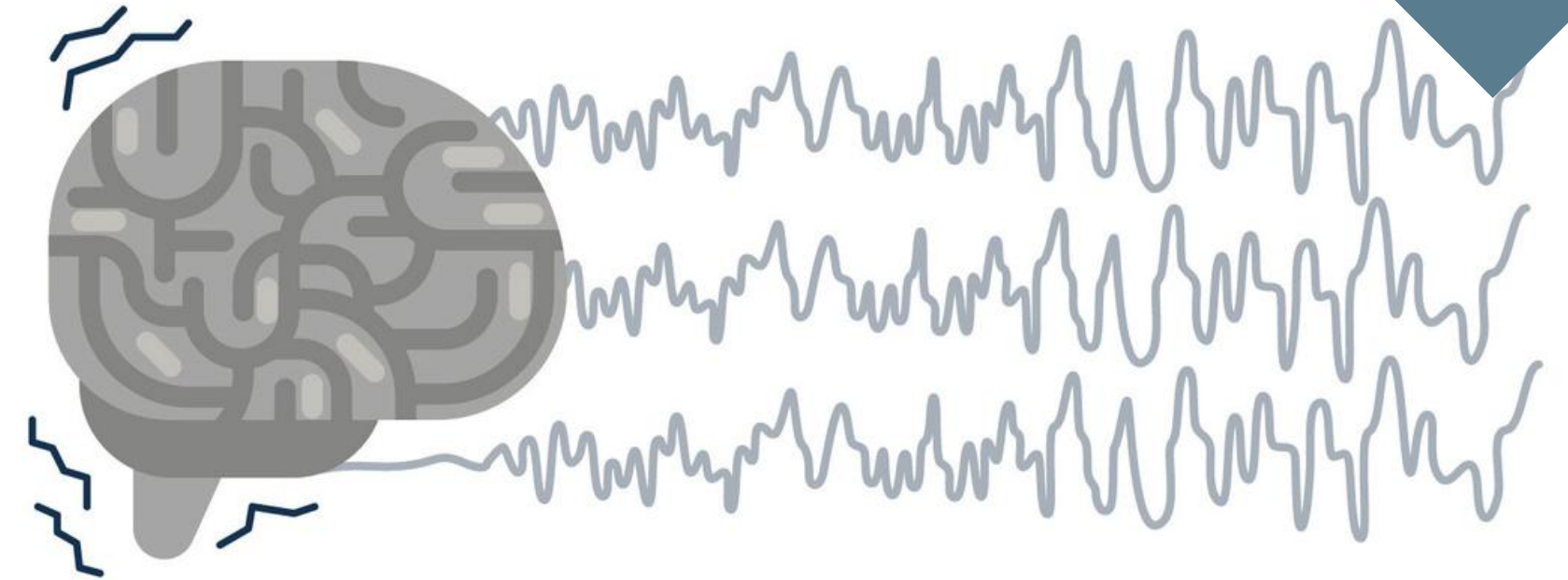
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1. PROBLEM DEFINITION

1.1. EPILEPSY

- A chronic neurological disorder.
- Characterized by recurrent, unpredictable seizures.



The cause of
epilepsy for
50%
of people
remains unknown



800,000

Australians will develop
epilepsy during their
lifetime

Around the world
an estimated
65 million
people have epilepsy
at any given time

1 in 3

people with epilepsy
will not gain full
seizure control with
medication



Source: WHO (2023); Global Burden of Disease Study 2019.

1.2. Effects of Epilepsy

- Sudden loss of consciousness.
- High risk of injuries and accidents.
- Constant fear and anxiety.

1.3. Why Is It Dangerous?

- Seizures can happen anytime, anywhere.
- Risk of SUDEP, especially during sleep.



2.MARKET STUDY

1.1. MARKET STUDY

Device	Company	Technology	Detection	Prediction	Accuracy	Key Feature
Embrace2 /EpiMonitor	Empatica	EDA, ECG, Motion, Temp	✔️ (Tonic-Clonic)	❌	85-90%	FDA Cleared, Low False Alarms
Smartwatches	Apple, Samsung	Heart Rate, Motion	⚠️ Limited (Abnormal Episodes)	❌	70-80%	Readily Available, High False Alarms
NightWatch	LivAssured	Heart Rate, Motion (Armband)	✔️ (Nocturnal Tonic-Clonic)	❌	80-85%	Optimized for Night Use, CE Marked
Seer Sense	Seer Medical	cEEG (Behind-ear), ECG	✔️ (Generalized)	❌	85-90%	Clinical Grade EEG, CE Marked
NeuroPace RNS System	NeuroPace	Intracranial EEG	✔️ (Focal Seizures)	✔️ (Prevents Seizures)	90% reduction	Only FDA-approved device for prediction & prevention
Research Prototypes	Universities / Labs	Multimodal + AI	✔️	✔️ (AI-based)	75-85%	In Development, Not Approved
BioSticker / BioStamp	BioIntelliSense	ECG, Respiration, Motion	⚠️ Limited (General Physiology)	❌	65-75%	FDA-cleared (General Use), Not Seizure-Specific

“NeuroPace RNS is an implantable clinical device and not comparable to wearable consumer solutions.”

1.2. MARKET GAP

Despite recent advances in wearable technologies:

- Most commercial devices are limited to seizure detection after onset.
- To date, no non-invasive wearable solution provides reliable early seizure prediction.

Furthermore, existing solutions lack full integration between:

- AI-based prediction.
- real-time mobile alerts.

Additionally, most solutions rely on either single-signal detection or invasive EEG systems.

1.3. OUR SOLUTION

- Develop a **wearable, non-invasive** system for continuous seizure monitoring Without usage of **complex signals (eg. EEG signals)**.
- Enable **early seizure prediction** using AI and multimodal physiological signals.
- Provide **real-time alerts** to patients, caregivers, and clinicians
- Improve **patient safety, independence, and quality of life.**



1. CLINICAL CONSULTATION & EXPERT VALIDATION

- Consulted neurologists and epilepsy specialists

Nirmeen Adel Kishk,MD

- Professor of Neurology ,Cairo University
- Moderator of CUEU
- Board member of the executive committee of ILAE-EMR (International League Against Epilepsy – Eastern Mediterranean Region)
- Vice secretary of ILAE-Egyptian chapter

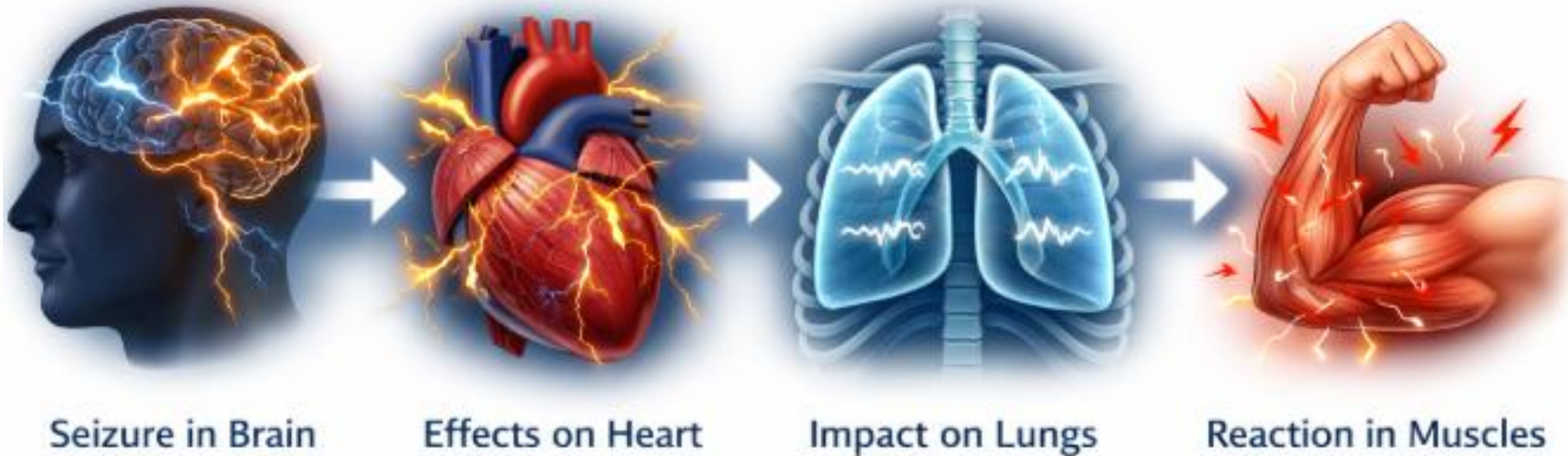
Khaled Hussein Salah Mughazy

- Neurologist- Cairo University Hospitals.

- Discussed current challenges in seizure monitoring and prediction
- Feedback confirmed the need for non-invasive, early-warning solutions suitable for daily use

1. SEIZURE-RELATED PHYSIOLOGICAL SIGNALS

- System is designed to acquire physiological signals associated with seizure prediction, What are the Main Physiological Signals?



WHY NOT EEG?

ENGINEERING & CLINICAL JUSTIFICATION

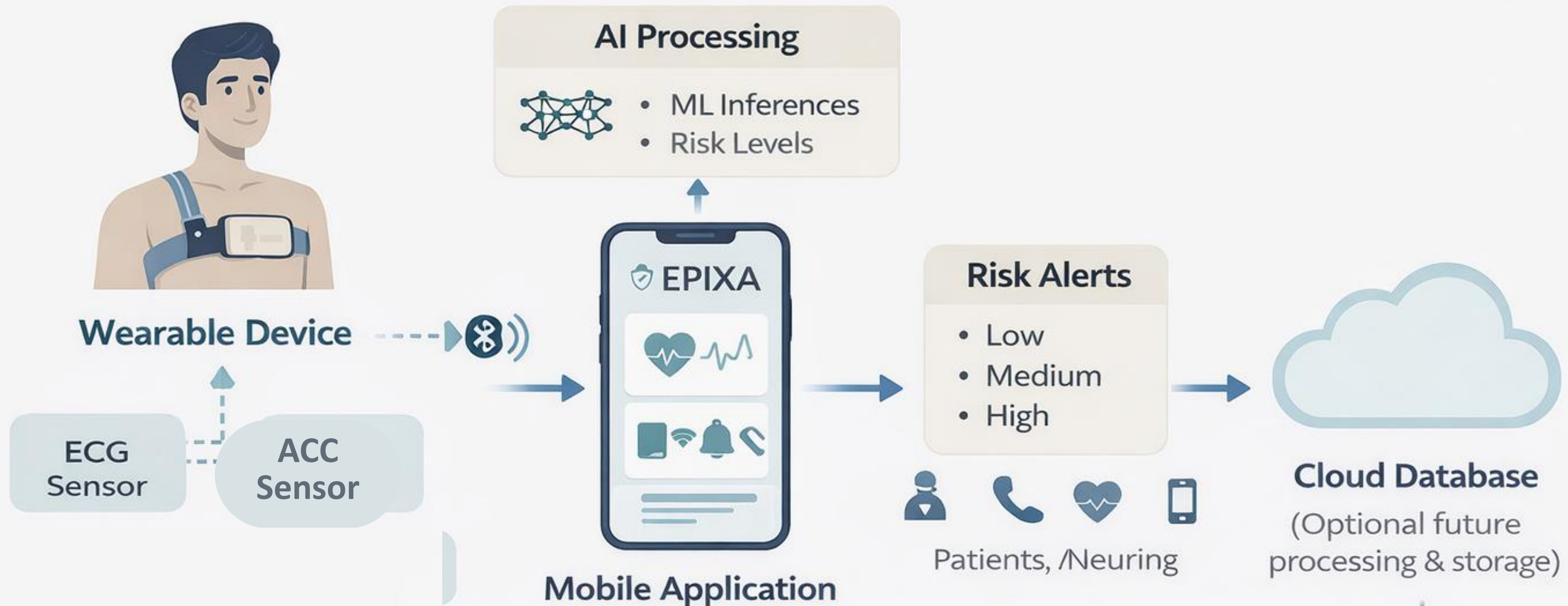
1. EEG is Gold Standard:

- Requires multi-electrode scalp setup
- Low patient compliance for daily wear
- High power consumption
- Motion artifacts in ambulatory use
- Expensive & clinically supervised

1. Our Engineering Rationale

- ECG captures autonomic pre-ictal changes
- ACC provides motion context
- Fully wearable & daily-life compatible
- Designed for long-term home monitoring

1. SYSTEM ARCHITECTURE

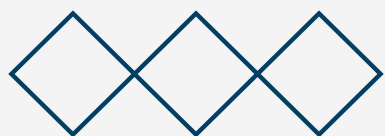


The system focuses on early risk estimation rather than direct medical intervention.

7.HARDWARE DEVICE

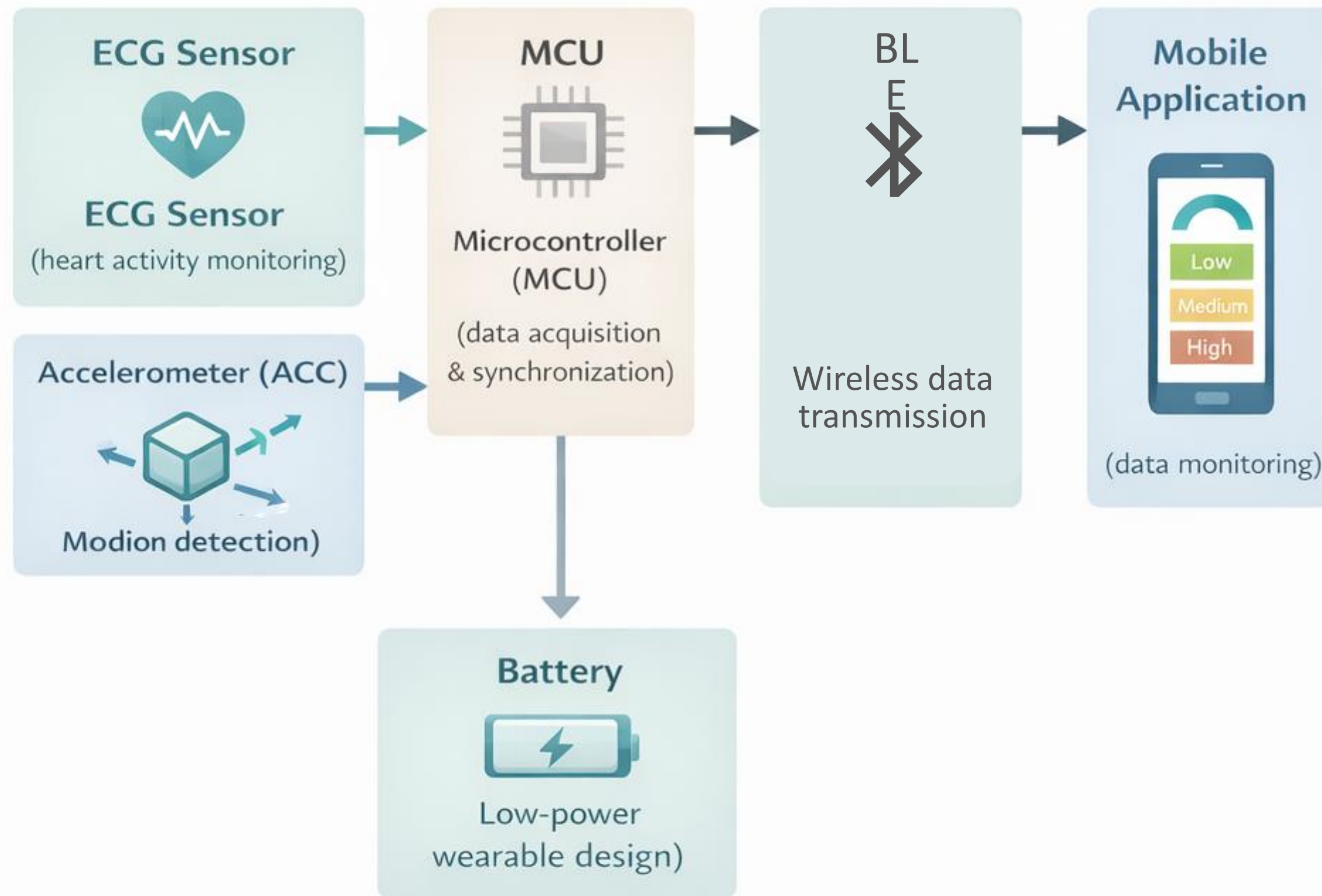
1. Design Constraints & System Requirements

- Non-invasive, daily-wear architecture
- Energy-efficient wearable design
- Low-latency, real-time risk estimation
- Personalizable AI for inter-patient variability



1. HARDWARE DEVICE

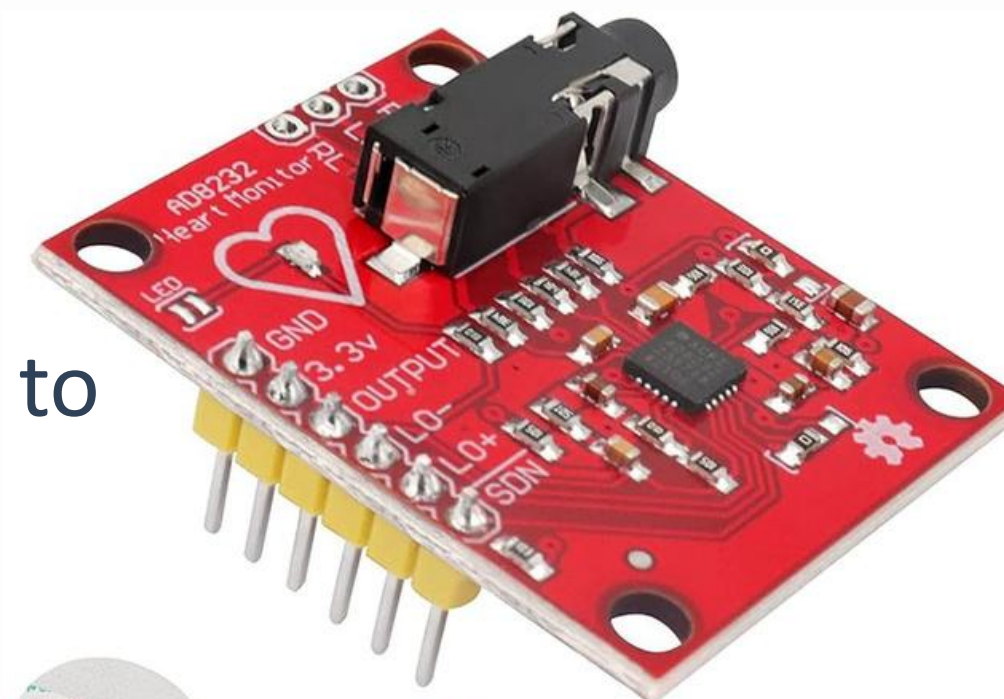
Circuit Diagram



1. HARDWARE DEVICE

Components

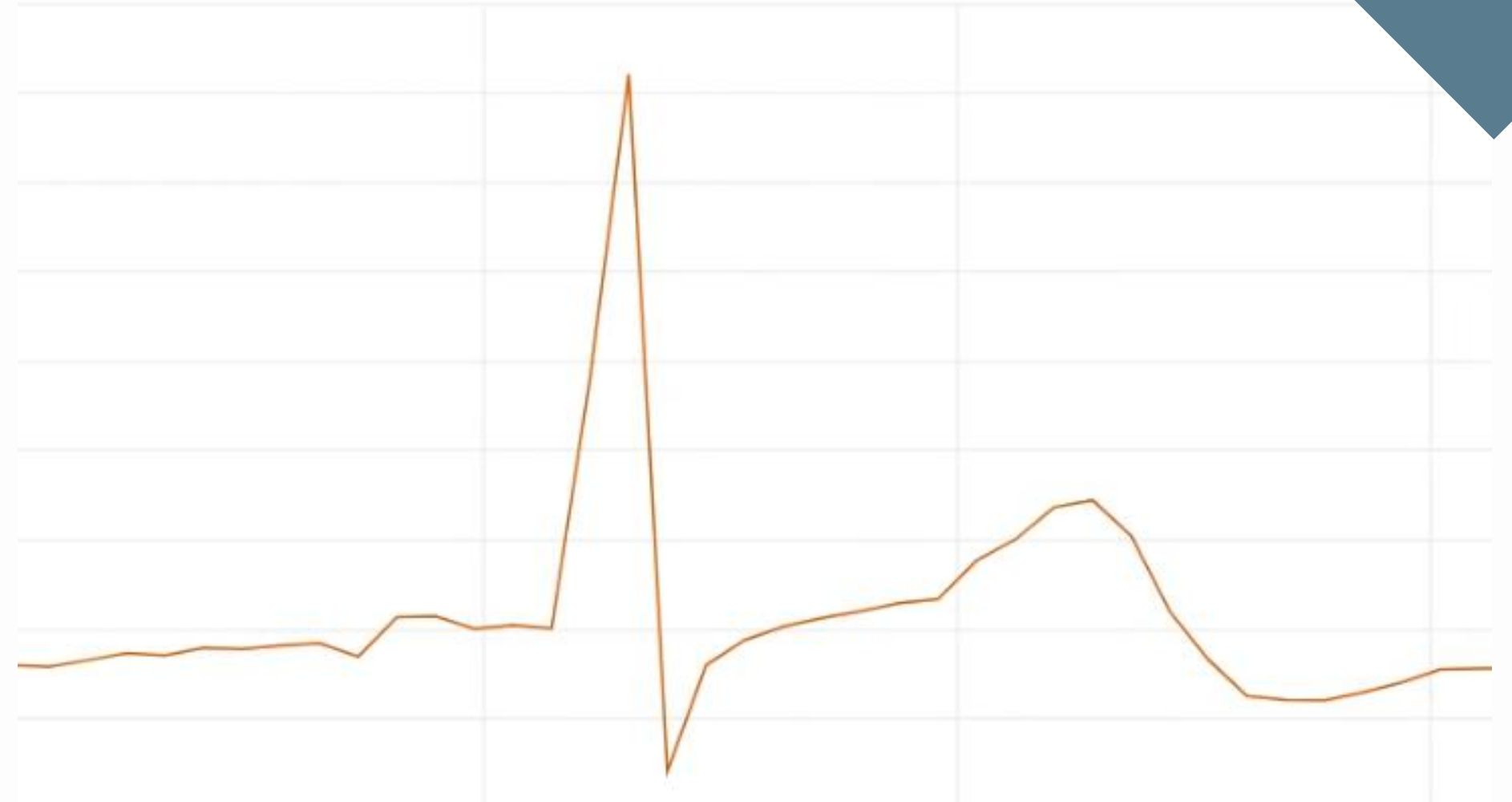
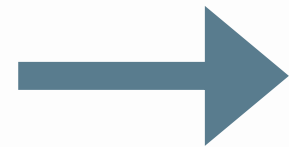
- ESP32 microcontroller => to collect the signals and send them via Bluetooth to mobile application.
- ECG sensor (AD8232) => three electrodes connected to chest.
- ACC sensor (ADXL345) => attached on chest to detect body movement.
- Power source 5-7 volt



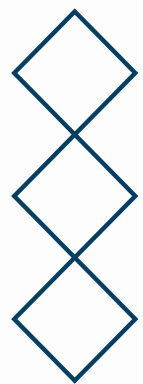
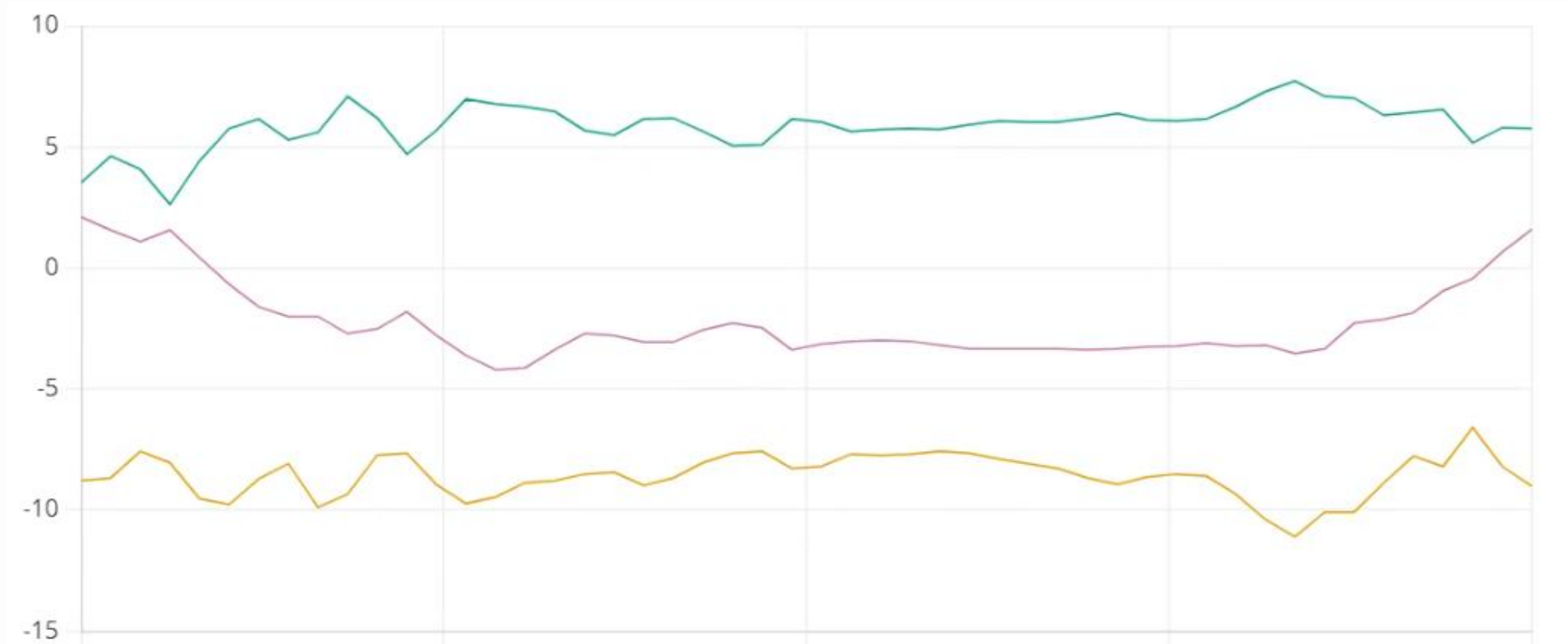
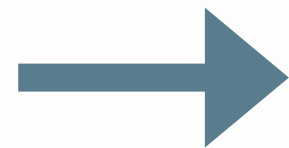
1. HARDWARE DEVICE

Sensors Output

- ECG Signal



- ACC Signal

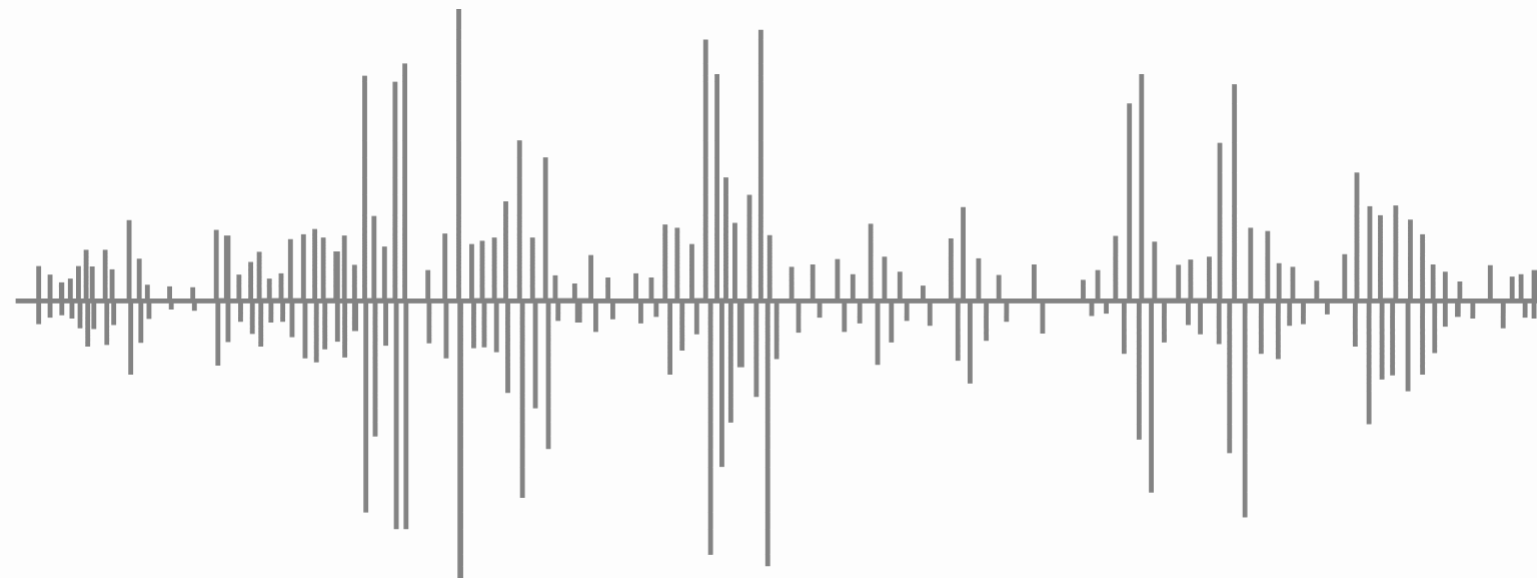


1. HARDWARE DEVICE

Sampling Frequency

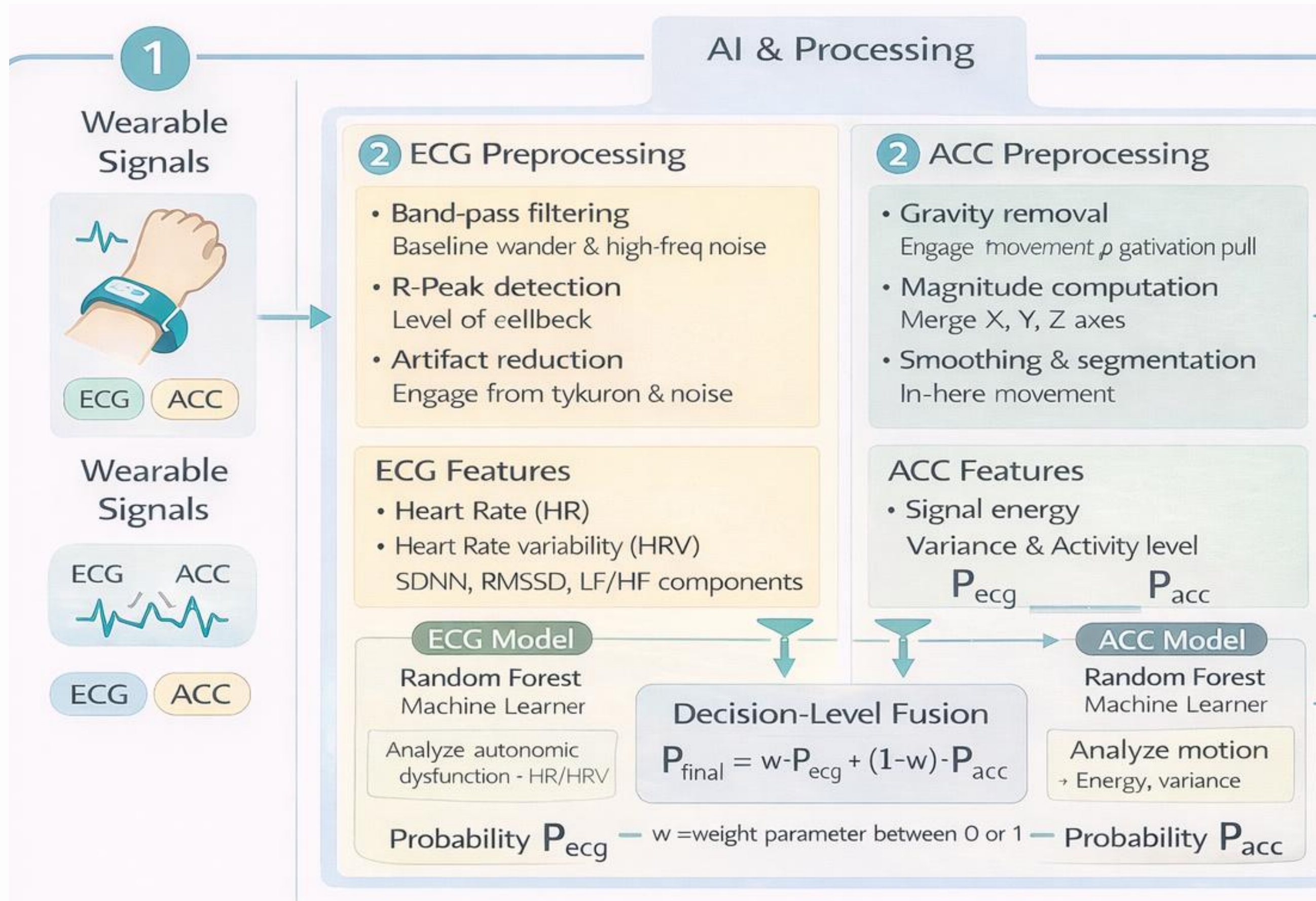
we have to re-sample each signal to meet the required frequency for model and then send the all the data mobile app using Bluetooth connection

- ECG 256 HZ
- ACC 25 HZ



1. PREDICTION MODEL

1.1. Architecture



1.2. SEIZEIT2 DATASET

- **Purpose:** First open dataset from a wearable device for focal epilepsy patients.
- **Scope:** >11,000 hours of multimodal recordings (EEG, ECG, EMG, motion sensors).
- **Participants:** 125 patients (41% female).
- **Seizures Recorded:** 886 focal seizures (Avg. duration: 58s, range: 3s – 16min).
- **Types:** Focal aware (317), Impaired awareness (393), Focal-to-bilateral tonic-clonic (55).
- **Seizure Onset:** Left 44%, Right 12%, Bilateral 1%, Unclear 43%.
- **Brain Regions:** Mainly temporal lobe (30%).
- **Data Format:** Organized in BIDS standard for research use.

1.3. DATA PREPARATION & EXPERIMENTAL SETUP

1. Preprocessing

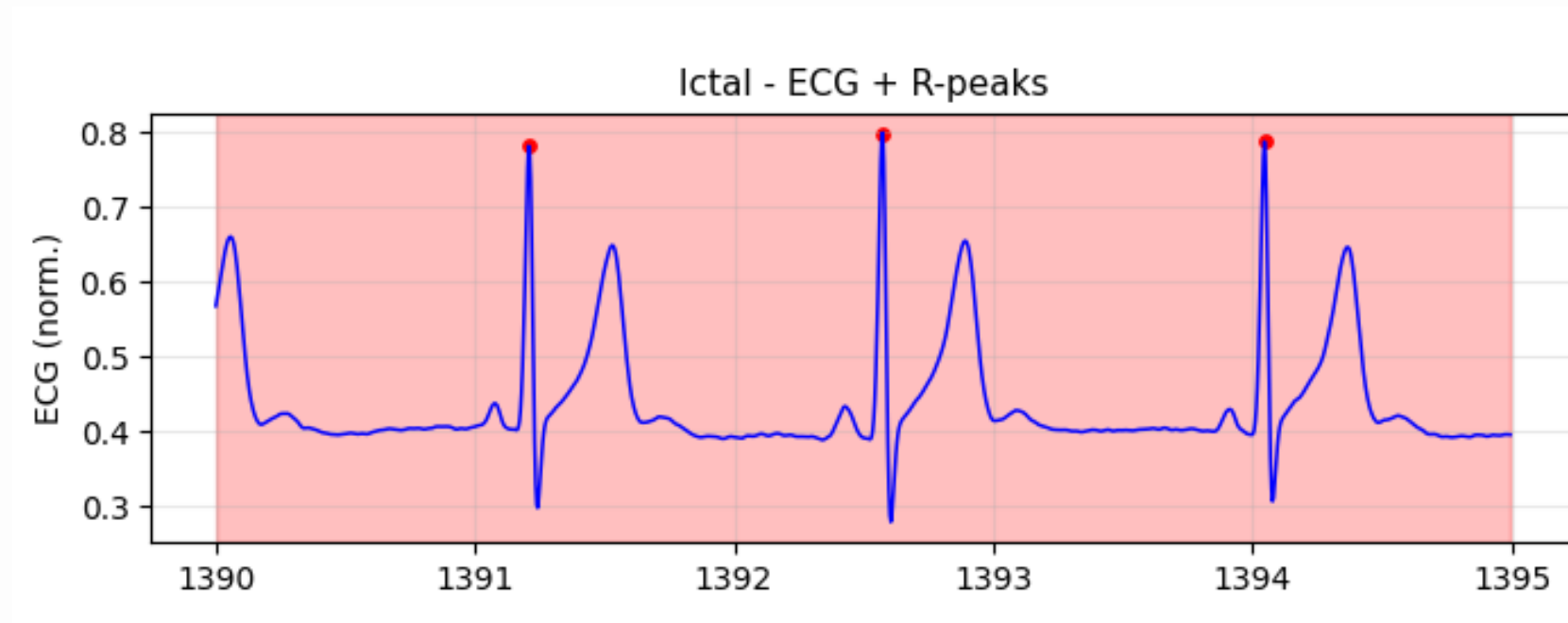
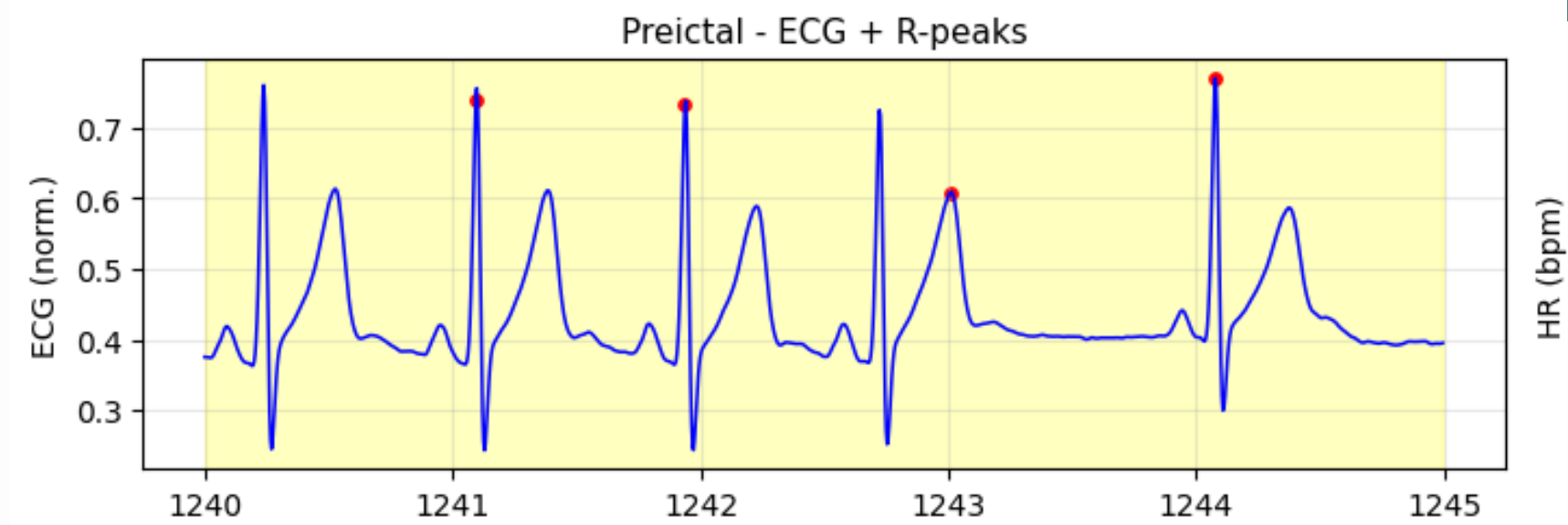
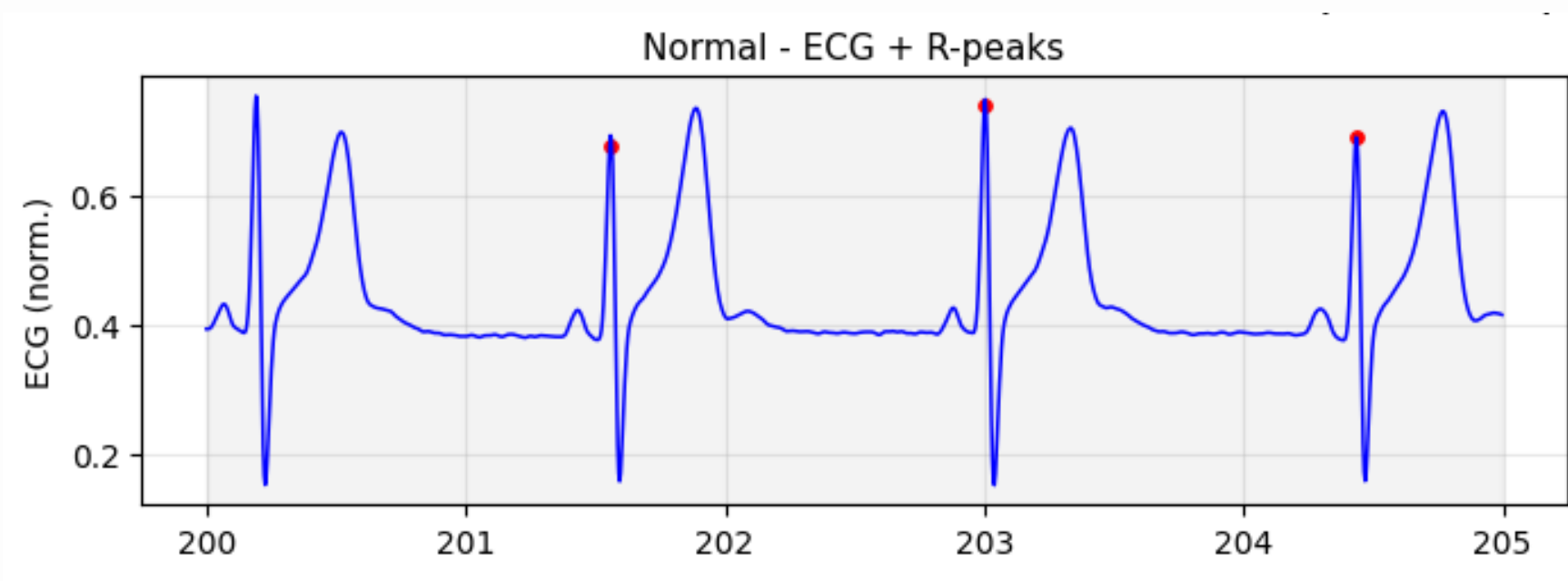
- Data cleaning
- Signals cleaning
- Normalization

1. Segment windows

- Divides ECG/ACC signals into fixed-length(60s) overlapping windows(10s)
- Assigns window-level labels (preictal, ictal, normal) with associated metadata for model training.

1. Prediction setup

- **Pre-ictal window:** 10 minutes



1.3. DATA PREPARATION & EXPERIMENTAL SETUP

1. Train/Test Strategy

Test Set:

- One completely unseen patient
- Used only for final model evaluation

Train + Validation Set:

- Remaining subjects only
- Split performed at subject level (train (80), valid (20%))

1. Class Imbalance Handling

- **Controlled under-sampling:** The majority class was reduced to a controlled 3:1 ratio relative to minority classes to mitigate class dominance while preserving representative samples.
- **Borderline-SMOTE:** Synthetic minority samples were generated near decision boundaries to enhance classifier sensitivity toward underrepresented seizure states

1.4. ECG FEATURES

ECG features capture early autonomic and electrophysiological changes that occur before seizure onset, enabling reliable early prediction of pre-ictal state

Feature	Normal Mean	Pre-ictal Mean	H-stat	Strength
maxHR	87.5	84.6	161.04	Very
Delta_HR	6.4	17	157.27	Very
stdHR	4.8	5.4	146.06	Very
meanHR	71	87	144.81	Very
meanHRV_bpm	71	66.5	143.33	High
n_beats	70.6	86.3	137.69	High
SDNN_ms	81.9	75.4	79.97	Medium
pNN50	7.2	11.2	57.64	Medium
minHR	61.9	63.5	33.88	Low
RMSSD_ms	81.9	74.3	32.21	Low



1.5. ACC FEATURES

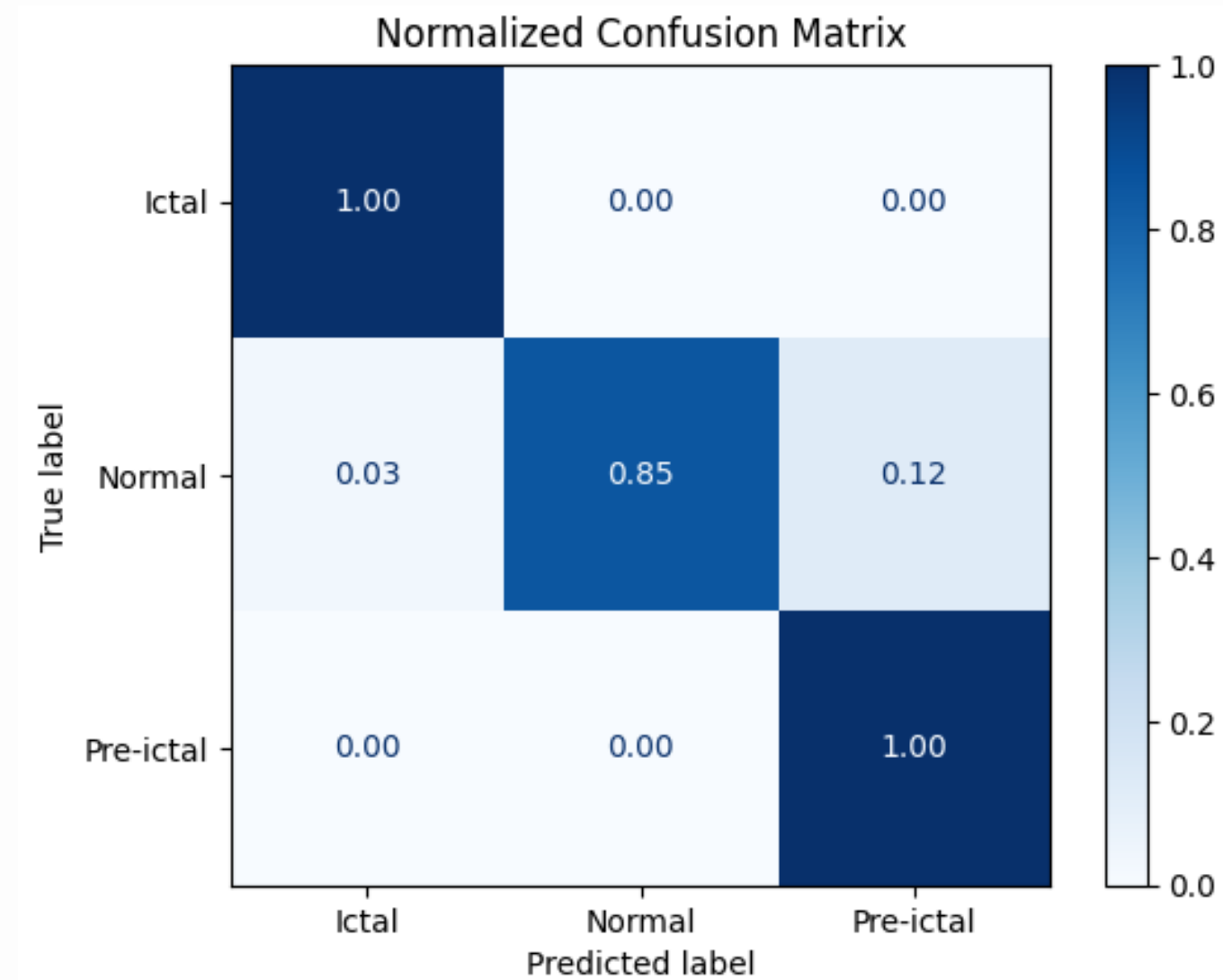
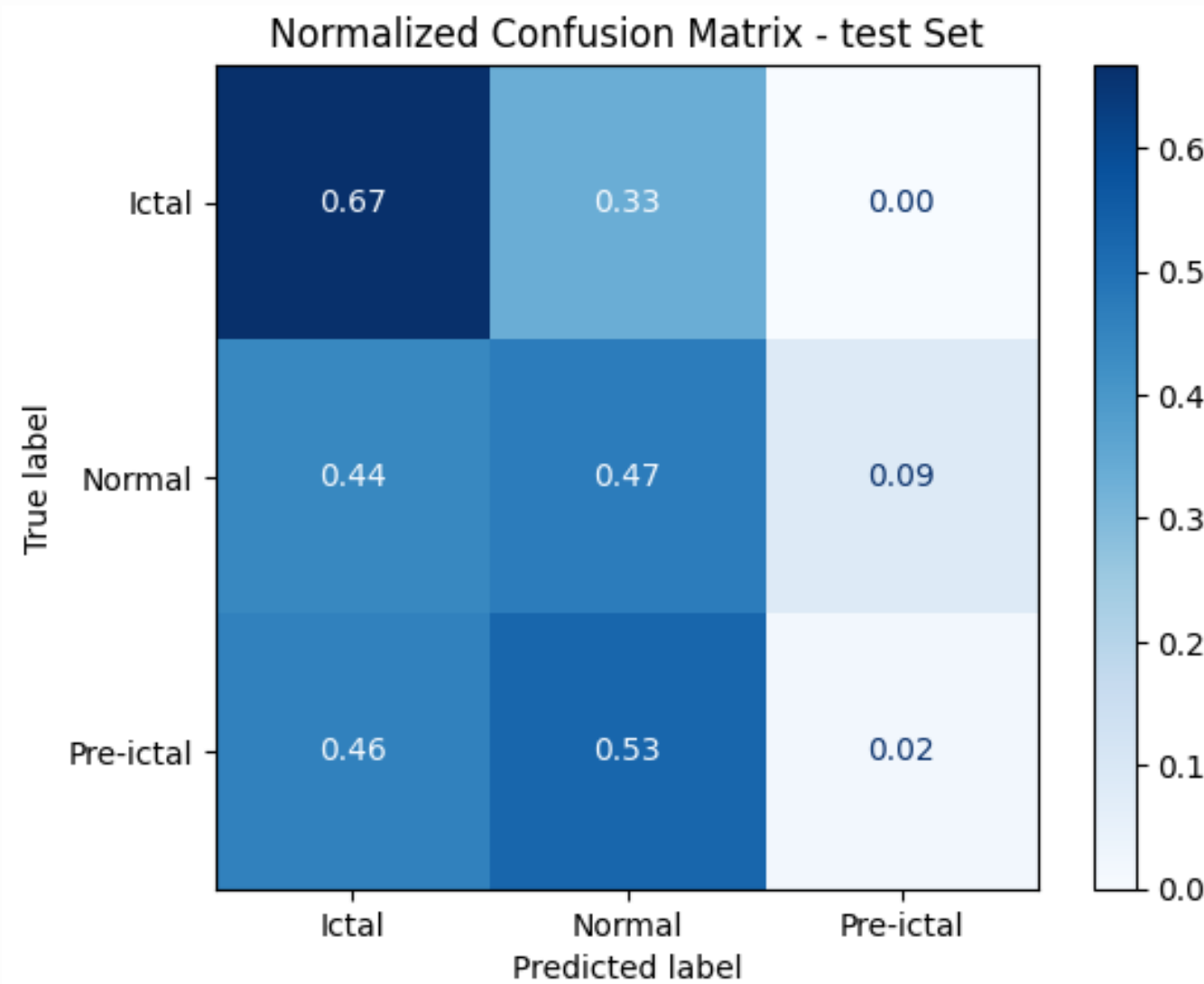
ACC features provide motion context by quantifying activity intensity and variability, helping distinguish seizure-related movements from normal daily activities and reducing false alarms.

Feature	Normal Mean	Pre-ictal Mean	H-stat	Insight
std_mag	0.00724	0.00837	1527.74	Strong n variabili
activity_score	0.84777	0.86437	472.64	↑ activit
rms_mag	0.84777	0.86437	472.64	↑ motion

1.6. Model Evaluation:

GENERAL VS PATIENT-SPECIFIC

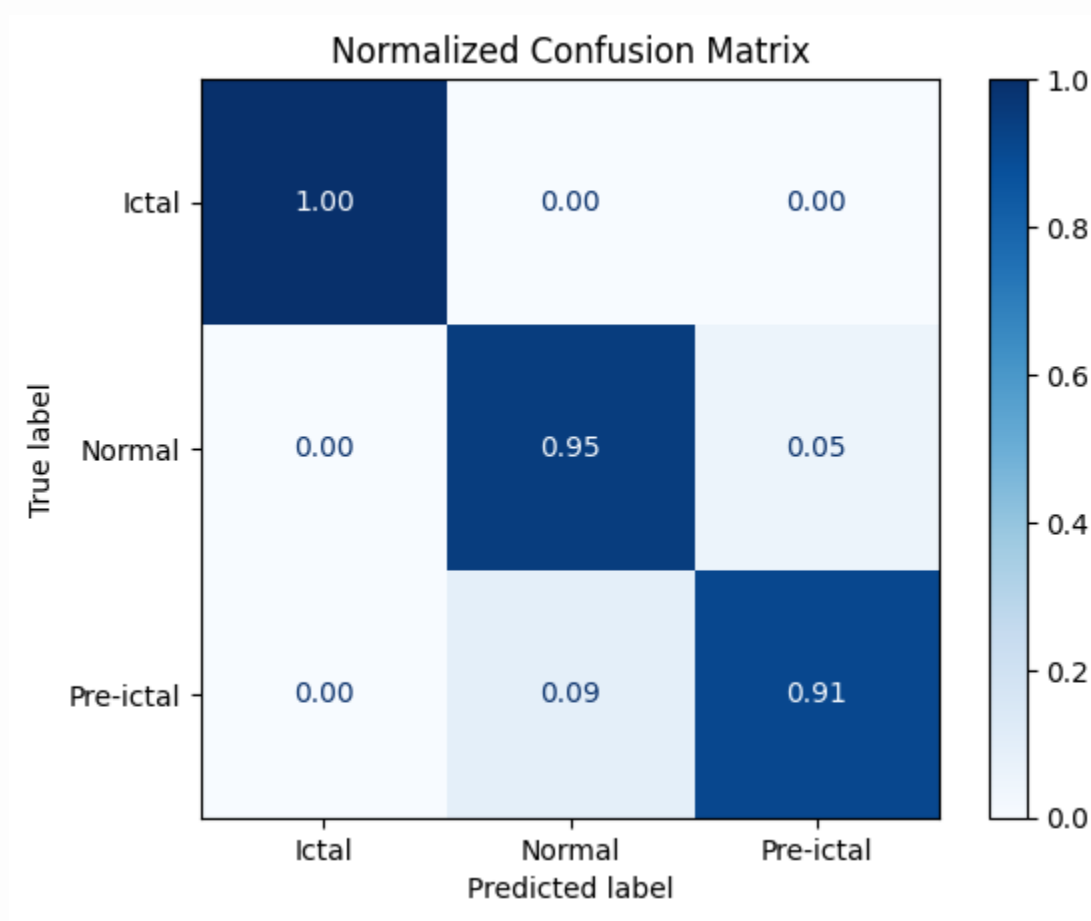
Patient-specific training improves reliability, while general models need personalization to handle inter-patient variability.



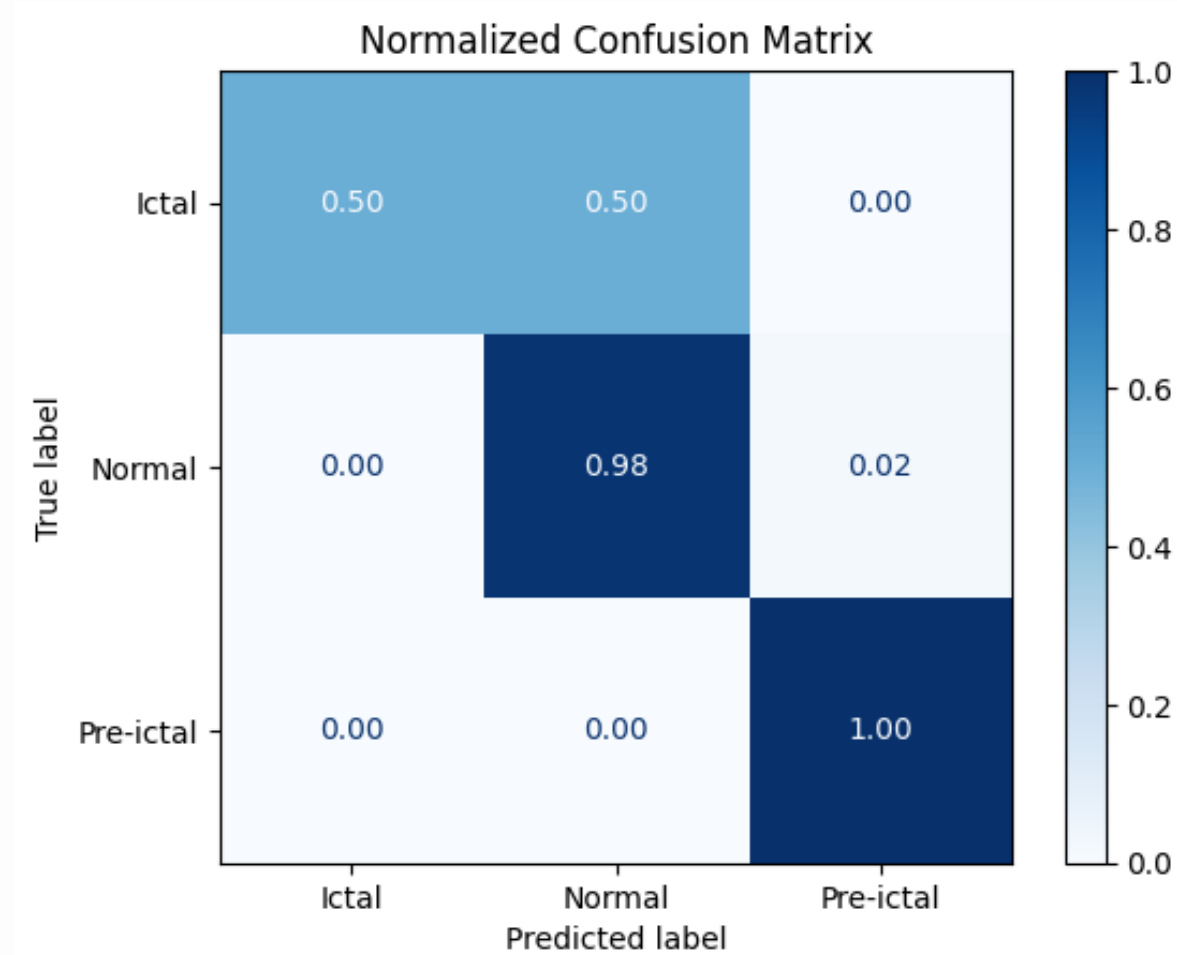
1.7. Model evaluation

The system currently performs best in patient-specific training settings, which is consistent with epilepsy literature.

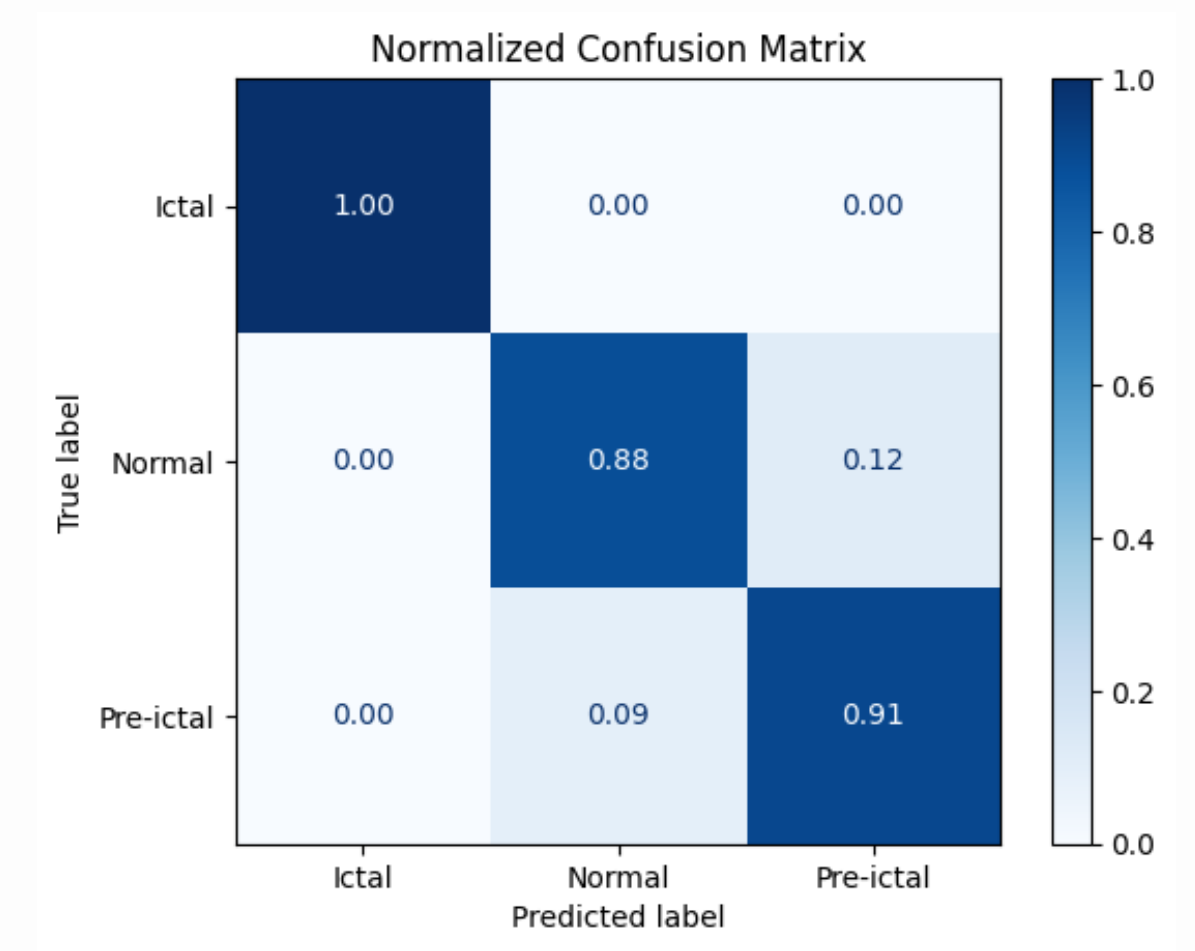
1



2



3



1.8. Fusion ECG&ACC

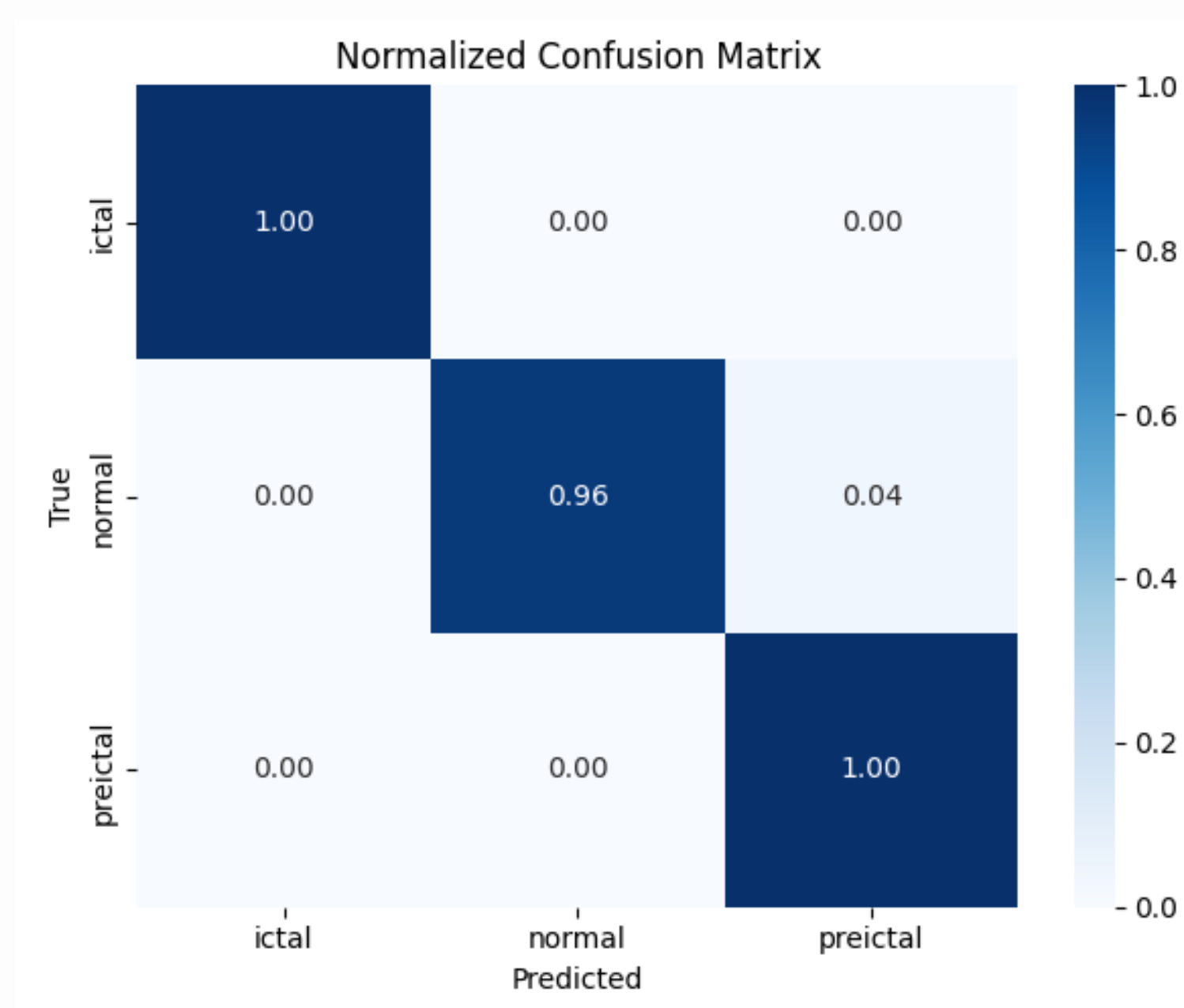
Fused probabilities are calculated as a weighted sum of ECG and ACC predictions, with and reflecting the relative importance of each source. The final prediction for each window is the class with the highest fused probability, integrating both signals for more reliable preictal detection.

$$\text{Prob fused} = 0.6 \times \text{Prob ECG} + 0.4 \times \text{Prob ACC}$$

Accuracy: 0.9744

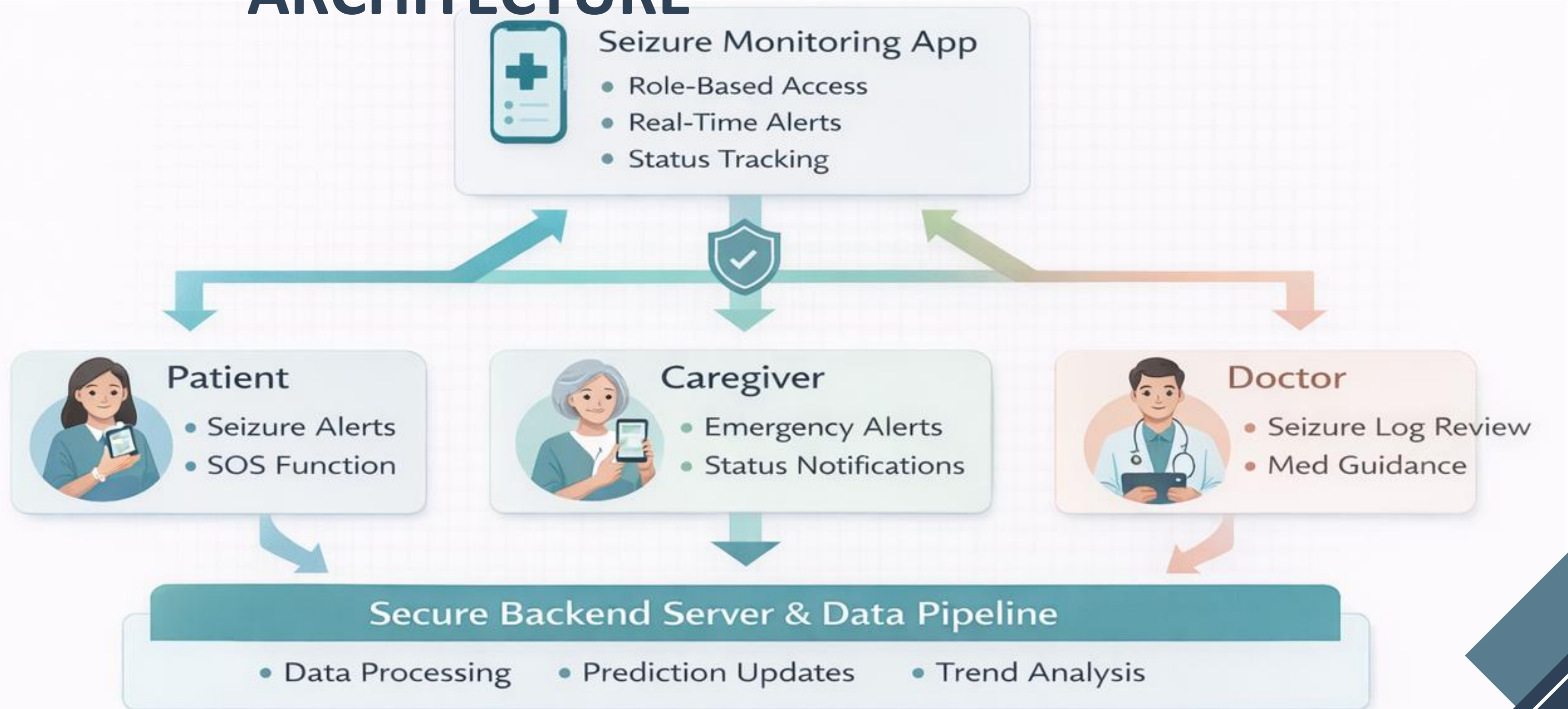
Classification Report:

	precision	recall
0	1.0000	1.0000
1	1.0000	0.9615
2	0.9167	1.0000



1. MOBILE APPLICATION

1.1. SYSTEM ARCHITECTURE



1.2. UI/UX LAYER

HUMAN INTERFACE DESIGN

- Translates AI predictions into patient-centered action
- Role-based access (Patient – Caregiver – Doctor)
- Interface layer between prediction and real-life response

The image shows a mobile application interface for a patient data entry form. The screen is titled "Fill your data" and features a profile icon labeled "patient". Below the icon, there are several input fields for personal information: Name, ID, Phone, Address, and Doctor ID. Each field has a corresponding icon (person, ID card, phone, location pin, and stethoscope) and a placeholder text "Enter your [field name]". The form is styled with a light blue background and rounded corners. At the bottom, there is a dark blue button labeled "DONE" and a circular profile picture placeholder.

1.3. CLINICAL & PSYCHOLOGICAL IMPACT



Patient Impact

- Anxiety Reduction
- Early Awareness
- Continuous Status Visibility



Caregiver Impact

- Real-Time Alerts
- Emergency Notification



Clinical Impact

- Trend Analysis
- Data-driven Treatment

1.4. EMERGENCY & SAFETY LAYER

Real-Time Seizure Alerts



- Immediate warning
- Sound and vibration
- Monitoring alerts

One-Tap SOS



- Patient instant call for help
- Continuous location tracking

SUDEP Risk Awareness



- Identify highrisk periods
- Educate patient & caregiver

SCIENTIFIC POSITIONING & PROJECT BOUNDARIES

1. System Classification

- AI-based seizure risk estimation system
- Non-invasive wearable research prototype
- Decision-support tool

1. What It Provides

- Pre-ictal risk probability
- Multimodal physiological fusion
- Real-time alerting layer

1. Clinical Boundaries

- Not a deterministic seizure predictor
- Not a replacement for clinical EEG monitoring

We estimate likelihood, not certainty.

1.TIMELINE



Current Focus: System Integration & End-to-End Prototype Development

1. NEXT STEP

AI Model Enhancement & Reliability Optimization

- Model refinement & hyperparameter tuning
- False alarm reduction
- Threshold calibration & validation

Objective: Improve clinical reliability and reduce false positives.

Backend & Mobile Application Development

- Backend architecture & secure database
- Real-time API communication
- Mobile application implementation
- AI-powered Clinical Decision Support Chatbot

Objective: Enable secure and intelligent continuous monitoring.

Hardware Prototype Finalization

- Wearable refinement & calibration
- Power optimization
- Stable MCU–Bluetooth communication

Objective: Deliver a reliable wearable prototype.

Full System Integration

- AI inference integration
- Wearable-to-mobile connectivity
- Backend synchronization
- End-to-end testing

Objective: Achieve a fully integrated working system.

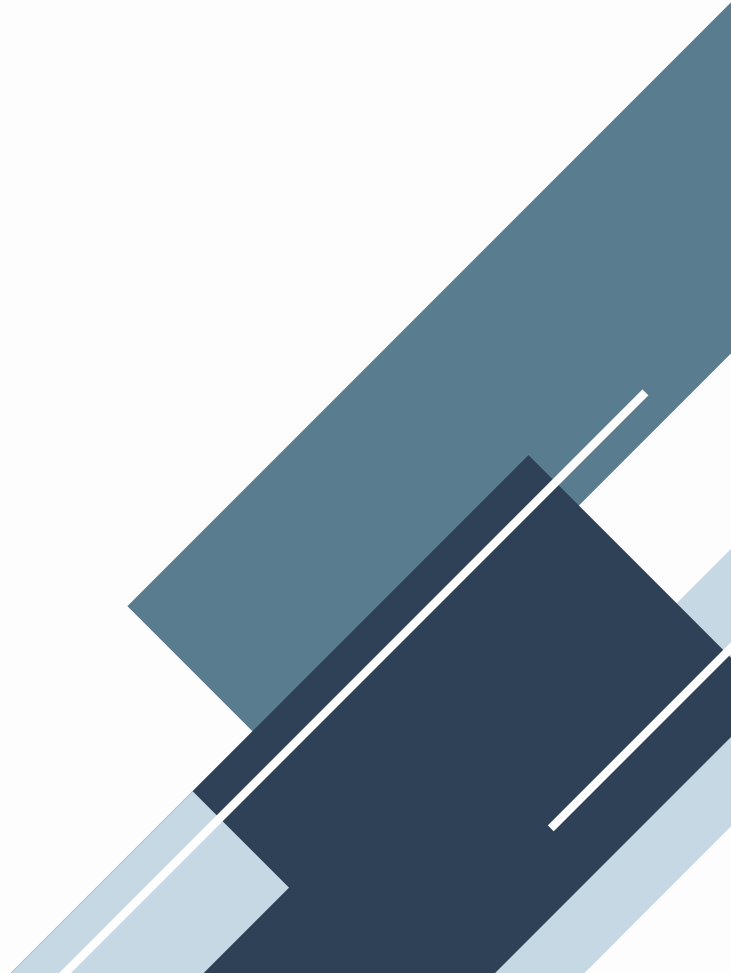
Monitoring Expansion

- Additional physiological sensors
- Multi-signal fusion validation

Objective: Enhance monitoring robustness.

1. REFERENCES

- https://www.mdpi.com/1424-8220/25/24/7687?utm_source=ecg-based-prediction-of-epileptic-seizures-using-machine-learning-methods
- [~~ECG-Based Prediction of Epileptic Seizures Using Machine Learning Methods~~ | IEEE Conference Publication | IEEE Xplore](#)
- [~~A novel wearable device for automated real-time detection of epileptic seizures~~ | BMC Biomedical Engineering | Full Text](#)
- [~~Wearable Ring-Shaped Biomedical Device for Physiological Monitoring through Finger-Based Acquisition of Electrocardiographic, Photoplethysmographic, and Galvanic Skin Response Signals: Design and Preliminary Measurements~~](#)
- <https://openneuro.org/datasets/ds005873/versions/1.1.0>



THANK you